



Lakshya JEE 2.0 (2024)

Limit, Continuity and Differentiability

DPP-02

1. $\lim_{x \rightarrow 0} \frac{\log(1+10x)}{x} =$
 (1) 10 (2) 1
 (3) 3 (4) 4
2. $\lim_{x \rightarrow 0} \frac{x \tan 2x - 2x \tan x}{(1 - \cos 2x)^2}$ is equal to
 (1) $\frac{1}{4}$ (2) 1
 (3) $\frac{1}{2}$ (4) $-\frac{1}{2}$
3. $\lim_{x \rightarrow 0} \frac{x \cdot 2^x - x}{1 - \cos x} =$
 (1) 0 (2) $\log 4$
 (3) $\log 2$ (4) None of these
4. $\lim_{x \rightarrow 2} \frac{2^x + 2^{3-x} - 6}{2^{-x/2} - 2^{1-x}}$ is equal to:
 (1) 8 (2) 4
 (3) 2 (4) None of these
5. $\lim_{x \rightarrow 1} \frac{4 - \sqrt{15x+1}}{2 - \sqrt{3x+1}}$ is equal to:
 (1) 2.5 (2) 3.5
 (3) 1.5 (4) none of these
6. $\lim_{x \rightarrow 2} \frac{3^x + 3^{3-x} - 12}{3^{-x/2} - 3^{1-x}}$ is equal to _____.
7. $\lim_{x \rightarrow 2} \left(\frac{\sqrt{1 - \cos\{2(x-2)\}}}{x-2} \right)$
 (1) Equals $\sqrt{2}$ (2) Equals $-\sqrt{2}$
 (3) Equals $\frac{1}{\sqrt{2}}$ (4) Does not exist
8. If $\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1} = \lim_{x \rightarrow k} \frac{x^3 - k^3}{x^2 - k^2}$, then k is:
 (1) $\frac{3}{2}$ (2) $\frac{8}{3}$
 (3) $\frac{4}{3}$ (4) $\frac{3}{8}$
9. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$ equals:
 (1) $\sqrt{2}$ (2) $2\sqrt{2}$
 (3) 4 (4) $4\sqrt{2}$
10. $\lim_{\alpha \rightarrow \pi/4} \frac{\sin \alpha - \cos \alpha}{\alpha - \frac{\pi}{4}} =$
 (1) $\sqrt{2}$ (2) $1/\sqrt{2}$
 (3) 1 (4) None of these
11. $\lim_{x \rightarrow \pi/2} (\sec \theta - \tan \theta) =$
 (1) 0 (2) $\frac{1}{2}$
 (3) 2 (4) ∞
12. $\lim_{x \rightarrow 0} \frac{x^4 (\cot^4 x - \cot^2 x + 1)}{(\tan^4 x - \tan^2 x + 1)}$ is equal to
 (1) 1 (2) 0
 (3) 2 (4) None of these
13. If $\lim_{x \rightarrow 0} \left\{ \frac{1}{x^8} \left(1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right) \right\} = 2^{-k}$ then the value of k is _____.



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (1)
2. (3)
3. (2)
4. (1)
5. (1)
6. (36)
7. (4)

8. (2)
9. (4)
10. (1)
11. (1)
12. (1)
13. (8)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>